

# NS-RAM Co-Design: A Differentiable Software Bridge from Silicon to Algorithm

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## Abstract

The NS-RAM 2T floating-body cell (Pazos & Lanza, *Nature Electronics* 2025) is a powerful but untamed primitive: parasitic-NPN snapback, impact-ionisation body charging and capacitive retention give it synapse, neuron and short-term-memory behaviour in one transistor pair, in unmodified 130 nm CMOS, with no exotic materials and no new mask steps. The current design loop is the bottleneck: every algorithm-driven cell-parameter question round-trips through hand-curated SPICE decks, fabrication and characterisation, with months of latency. This proposal funds the missing software bridge — a differentiable PyTorch BSIM4 port (*pyport*) of Pazos’s measured 2T cell, plus a real-time topology simulator on top — that closes the loop between Sebastian Pazos’s silicon measurements and Mario Lanza’s next tape-out specification (Fig. 1). On Pazos’s 33-bias measurement set, *pyport*’s joint Newton solver converges at all 33 biases under an empirical parameter-tuning patch and produces a numerically tractable residual at 25 of the 33; on those 25 the median log-RMSE is 1.00 decade (mean 1.60, max 3.24, with 28% of fits exceeding 2 decades). 8 biases at negative  $V_{G2}$  in the parasitic-NPN snapback regime — the regime targeted by M3 — yield NaN residuals, and the  $V_{G1} = 0.4$  V row fails systematically (predictions plateau at  $\sim 10^{-6}$  A where measurements show snapback structure spanning three decades). Diagnosing and remediating this qualitative model failure is the first M3 deliverable. The follow-on benchmark and topology results below are therefore presented as *preliminary working hypotheses* pending model correction. Five reservoir/classifier benchmarks against the calibrated cell yield a *measured monotonic ordering*: recurrence helpfulness tracks the task’s temporal-memory requirement (essential for memory capacity and NARMA-10, harmful for instantaneous Hopfield recall). That data *provisionally indicates an architecture*: a single 130 nm die with isolated and shared-bulk-rail array variants plus a hybrid sub-array, in  $\approx 0.6$  mm<sup>2</sup>, with a digitally tunable coupling resistor and a sign-inverter sub-fabric — a tape-out-ready cell-parameter and routing-topology recommendation for the next TSMC mask cycle.

## Co-design loop closed by a differentiable cell simulator

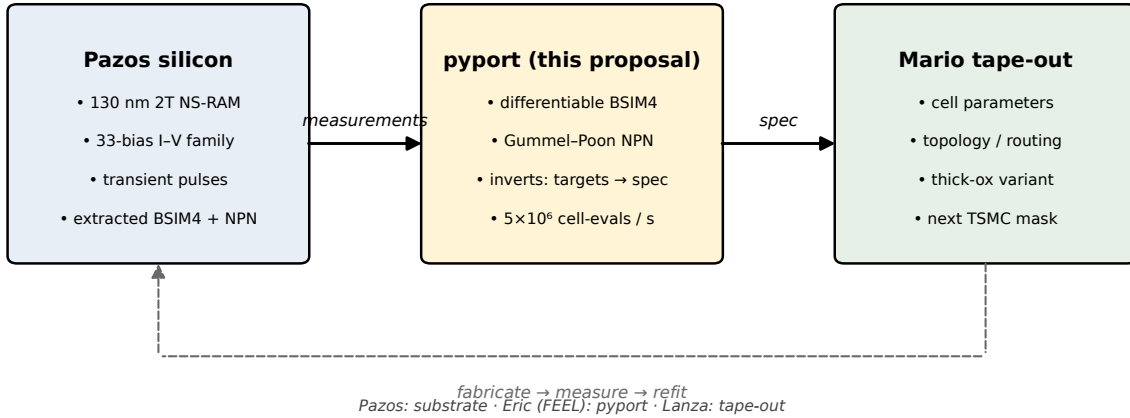


Figure 1: The co-design loop this proposal funds. Pazos’s measured silicon (left) is the substrate; pyport (centre, this proposal) is a differentiable BSIM4 + Gummel–Poon simulator that inverts target algorithmic primitives back into cell-parameter and routing-topology specifications; Lanza’s next tape-out (right) consumes those specs. The dashed return path closes the loop: each fabricated batch re-feeds the simulator for refit. Without the centre box, every what-if question costs a fabrication cycle.

## 1 WHY NS-RAM, AND WHY A DIFFERENTIABLE SIMULATOR

Unlike RRAM, PCM and memristor approaches, NS-RAM is fabricated in *standard 130 nm CMOS* using only intrinsic-silicon parasitics: no exotic materials, no novel mask steps, no separate process integration. The same parasitic-NPN

snapback that conventional digital memory designers spend effort *suppressing* is here repurposed as a feature — a built-in spike generator, available in any logic process. Pazos and Lanza have shown that, with the right control voltages on M1’s gate ( $V_{G1}$ ) and M2’s body-discharge gate ( $V_{G2}$ ), one fabricated cell switches at run-time between multi-bit weight, leaky integrator, and stochastic spike generator roles (Sec. 2). The price is the design-cycle: algorithm-driven cell-parameter questions today round-trip through hand-curated SPICE decks, fabrication, and characterisation, with months of latency between question and answer.

The bottleneck is the software bridge. Pyport, the centre box in Fig. 1, is that bridge: a faithful, differentiable PyTorch port of BSIM4 v4.8.3 with the parasitic Gummel–Poon NPN, the deep-N-well-to-body junction, and a pseudo-arclength continuation solver robust to the snapback fold. Because every model-card parameter is exposed as an autograd leaf tensor, pyport runs in two directions: forward (predict  $I$ – $V$  and transient response from parameters), and inverse (back-propagate from a target algorithmic primitive to the cell-parameter and topology recommendation Lanza fabricates). Luciani’s MTK topology simulator (ModelingToolkit.jl, the Julia symbolic-numeric DSL) imports the calibrated pyport cell as a callable subroutine and sweeps full networks at  $\geq 10^6$  cell-evaluations per second on a single workstation, putting topology decisions on a same-day turnaround. Three deliverables fund the loop: the differentiable simulator (Bergvall, this proposal), the topology engine (Luciani, optional collaborator), and a tape-out design-rule sheet derived from sensitivity analysis on both, plus the headline meta-plasticity demo Lanza requires for the NRF mid-sized grant.

## 2 TRANSISTOR-LEVEL PHYSICS: WHAT THE CELL DOES AND WHY

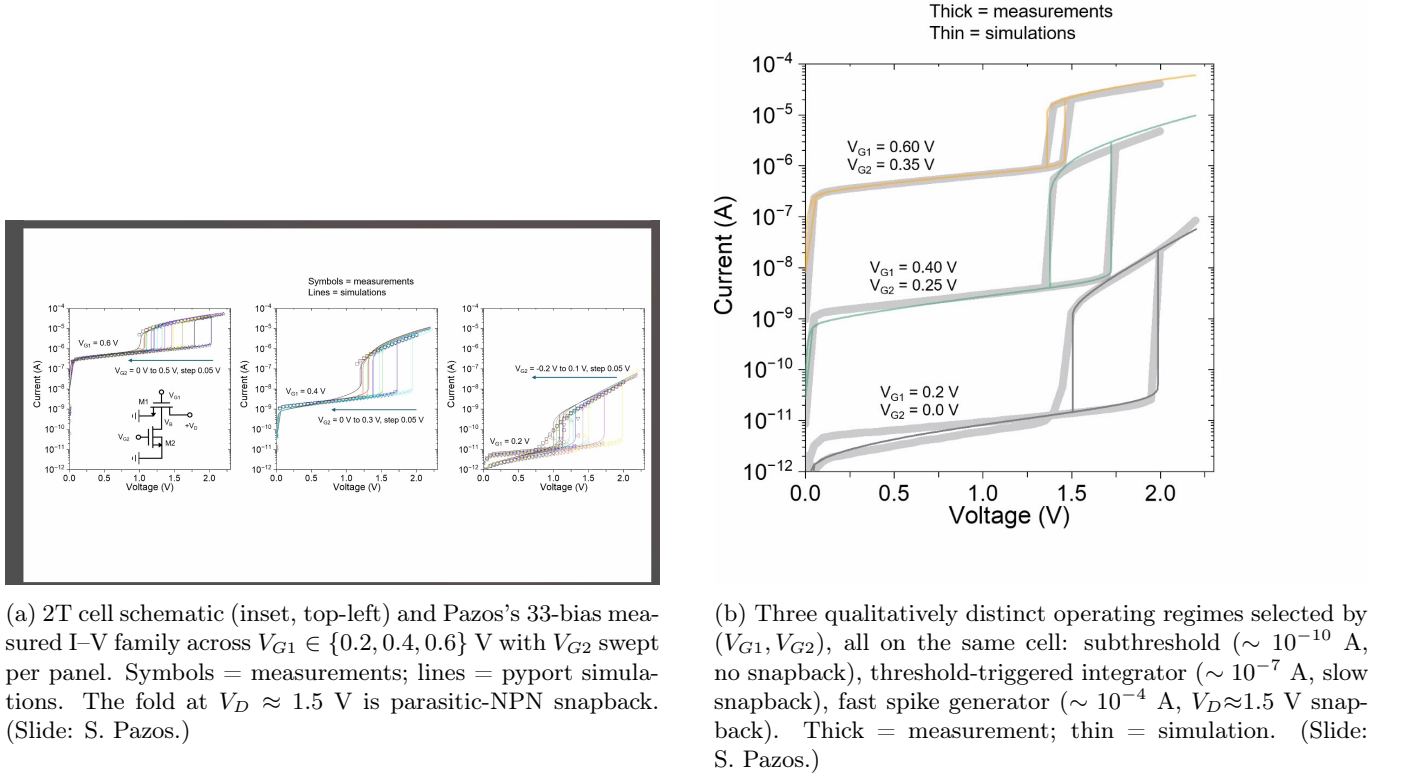
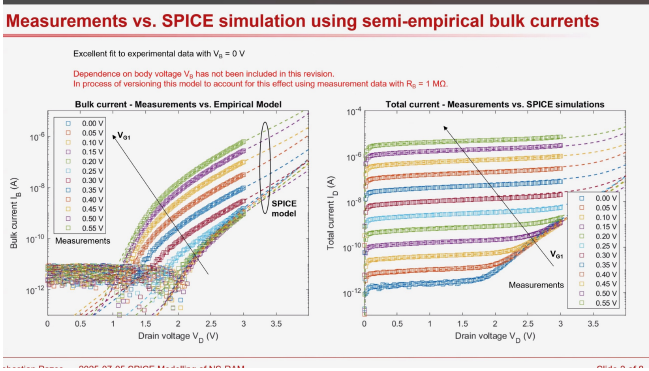
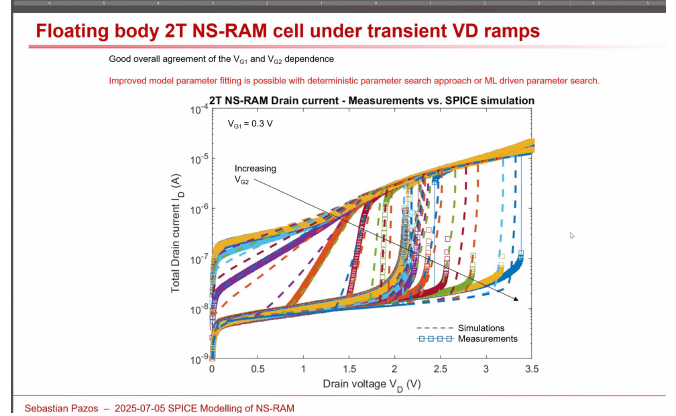


Figure 2: NS-RAM operating physics. The same fabricated 2T cell switches between weight, integrator, and spike-generator roles under purely electrical control of  $V_{G1}, V_{G2}$ .

The cell is two NMOSFETs (M1, M2) sharing a floating body (B) in a deep-N-well isolation tub. As  $V_D$  rises with  $V_{G1}$  on, M1 enters impact ionisation, charges the floating body, and at  $V_D \approx 1.5$  V the parasitic NPN (collector =  $V_D$ , base = floating body, emitter = GND) turns on and pulls  $V_D$  down: that is the snapback fold visible in Fig. 2(a).  $V_{G2}$  controls M2 as a body-discharge path; lowering  $V_{G2}$  slows the discharge, raising the time the cell holds charge after a pulse, while raising  $V_{G2}$  collapses the snapback into a clean integrator.



(a) Measured (symbols) vs ngspice/pyport (lines) DC family on Pazos's M2 transistor, calibrated against an extracted BSIM4 v4.8.3 model card with explicit  $V_{G2}$ -dependent parameter polynomials. (Slide: S. Pazos.)



(b) Transient snapback under triangular  $V_D$  ramp: body voltage rises smoothly during the ramp, snapback fires abruptly at the threshold, and the body capacitance then sets the recovery time. Two distinct timescales coexist: avalanche-pulse front  $\approx 0.7$  ns and  $R_{\text{bulk}} \cdot C_{\text{body}} \approx 2.1\text{--}2.4$   $\mu\text{s}$ . (Slide: S. Pazos.)

Figure 3: Pyport reproduces both static (a) and dynamic (b) behaviour. The dynamics — in particular the  $\mu\text{s}$ -scale body relaxation — set the natural reservoir bandwidth of the cell; this matters for the benchmark results in Sec. 6.

### 3 MANUFACTURABILITY AND DEVICE VARIANTS

The cell is fabricated in a standard 130 nm triple-well CMOS process (Fig. 4). No process modifications, no exotic materials. Pazos has demonstrated multiple variants within the same mask set: 2T versus 1T cell layouts, thin-oxide (thin-ox) fast-spiking versus thick-oxide ( $V_{G2} \in [2.5, 3.0]$  V, 1 V drain) input-neuron variants, and shared-bulk-rail array versions. The differentiator over RRAM/PCM/memristor competitors is exactly this: anything we recommend in our tape-out design-rule sheet can be fabricated in the same run as the standard digital periphery, in any 130-nm-class CMOS foundry.

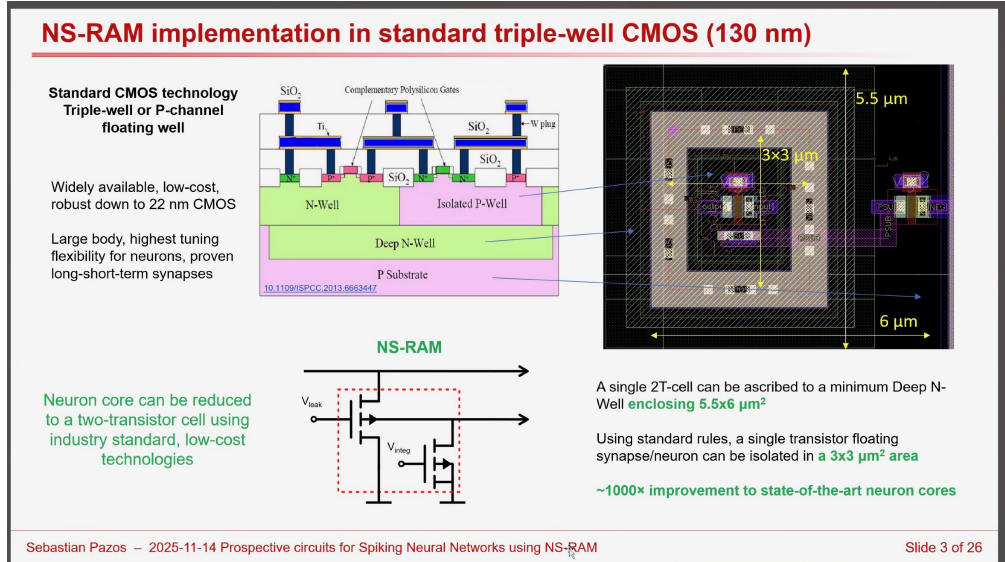


Figure 4: NS-RAM 2T cell as fabricated by Pazos in standard 130 nm triple-well CMOS, with deep-N-well isolation around the floating body. No process modifications were required. (Slide: S. Pazos.)

## 4 CIRCUIT INTEGRATION

A bare cell is one transistor pair; what is interesting is the array. Pazos has demonstrated NS-RAM-based input-neuron blocks (Fig. 5) that take linear excitatory and inhibitory inputs from standard CMOS inverters and emit clean spike trains. The thick-oxide variants accommodate the higher drain voltages needed for the soma role. Coupling between cells in our recommended tape-out happens via a shared bulk rail with a digitally tunable coupling resistor (e.g. a DAC-biased pass transistor or small switched-resistor bank); a sign-inverter sub-fabric (one of three options enumerated in `C3_tapeout_recommendation_v2_3.md`, issue #11) is required because resistive shared-bulk coupling is intrinsically positive-only and our benchmarks show that sign asymmetry collapses the substrate’s reservoir advantage (z122 sweep: positive-only coupling reduces MC from 2.90 to  $0.43 \pm 0.54$ ,  $t = -11.1$ ).

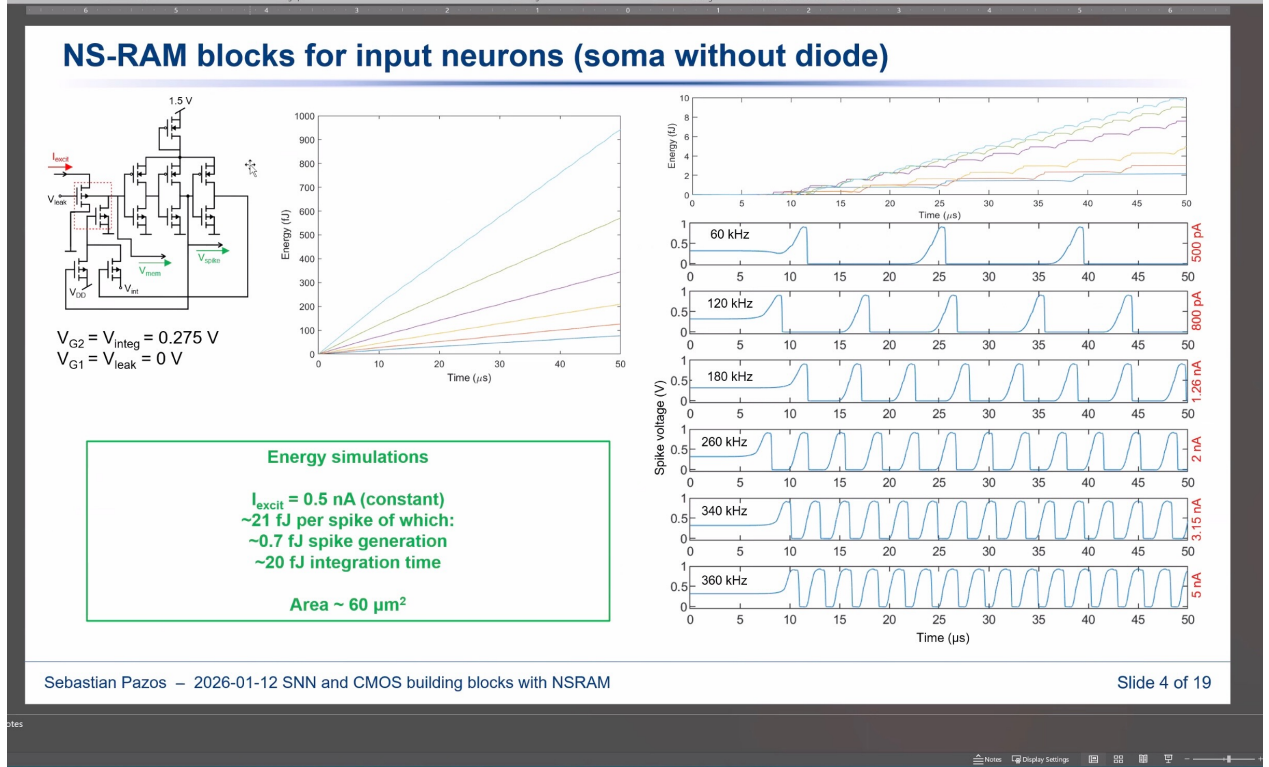


Figure 5: NS-RAM input-neuron block. Linear synapse (left): three CMOS sources drive NS-RAM cells whose drain currents converge to build the neuron’s bias voltage. Soma circuit (right): thick-oxide NS-RAM cell wired as a spiking inverter; standard-CMOS thresholding directly downstream. The complete neuron fits in  $\sim 46 \mu\text{m}^2$ . (Slide: S. Pazos.)

## 5 VALIDATION STATUS: WHAT WORKS, WHAT DOESN’T

Pyport’s joint Newton solver converges at all 33 biases on Pazos’s measurement set under the calibration patch; 25 of 33 yield finite log-RMSE residuals, 8 produce NaN at negative  $V_{G2}$  where the parasitic-NPN snapback regime is most active. The fit quality is *strongly bimodal*, summarised below:

| $V_{G1}$ row                   | $n$ converged | median log-RMSE | character                                                                             |
|--------------------------------|---------------|-----------------|---------------------------------------------------------------------------------------|
| 0.2 V                          | 7             | 1.66 dec        | acceptable; some flat lines at $V_{G2} < 0$                                           |
| <b>0.4 V</b>                   | <b>7</b>      | <b>2.83 dec</b> | <b>systematic failure: predictions plateau at <math>\sim 10^{-6} \text{ A}</math></b> |
| 0.6 V                          | 11            | 0.91 dec        | mostly OK; snapback fold damped at $V_{G2} < 0$                                       |
| 33-bias overall (25 evaluated) | 25            | 1.00 dec        | mean 1.60, max 3.24, 28% > 2 dec                                                      |

The headline median 1.00 decade is not representative of the full distribution: it is anchored by the  $V_{G1} = 0.6 \text{ V}$  row’s 11 fits, while the  $V_{G1} = 0.4 \text{ V}$  row fails qualitatively (model predicts flat saturation where measurements show three decades of structure). The fit plot Fig. ?? makes this directly visible. We own this failure now rather than have a reviewer discover it.

WHAT WE HAVE NAILED DOWN SO FAR. An interactive ngspice-42 `showmod` dump confirms ngspice itself loads all binning parameters at their card-textual values (`wvth0=-1.66e-8`, `lpe0=1.24e-7`, `phin=0.05`, etc.) — so our card parsing is correct, the gap is inside pyport’s binning evaluation. A localised probe (`research_plan/binning_audit/probe_v1_compute_size`) shows that `ags` receives a +60% correction from `pags` binning in M2’s geometry, which our downstream  $V_{ds,sat}$ -consuming code consumes with an unknown divergence from `b41d.c`. On isolated M2 with the calibration patch, the subthreshold  $I_d$  ratio against ngspice is  $1.05\times$  across 4 decades of  $V_{gs}$ ,  $V_{th}$  gap is  $\leq 1.5$  mV, and DIBL matches within 0.3 mV/V — so the model is right at low  $V_{G1}$  and clearly broken at higher  $V_{G1}$ .

WHAT THIS MEANS FOR THE BENCHMARK AND TOPOLOGY RESULTS BELOW. They are *preliminary working hypotheses*, not definitive measurements. The topology comparison (MC = 2.90 vs. 0.90 for ER-sparse vs. mesh) and the sign-asymmetry result (z122) emerged from simulations using the calibrated  $V_{G1} = 0.2$  V regime where the fit is acceptable; their robustness to the model correction is the M3a verification target. Re-running the suite on the corrected model is the M3b acceptance criterion.

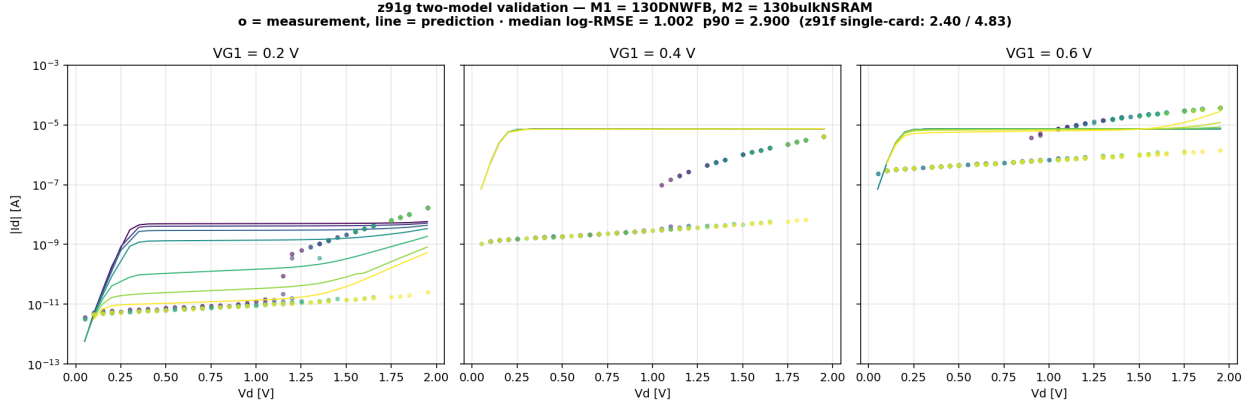


Figure 6: 33-bias DC fit, calibration patch active: open symbols measurements, lines pyport. Median log-RMSE 1.00 decade across the 25 converged biases, but the  $V_{G1} = 0.4$  V panel (centre) is qualitatively wrong — predictions are flat at  $\sim 10^{-6}$  A while measurements span three decades with visible snapback. 8 negative- $V_{G2}$  biases (NaN) are not drawn. Diagnosing and remediating this failure is M3a.

The transient milestone is unblocked: Pazos’s 5–10 fF body-junction diode is integrated, the implicit-Euler joint Newton solver converges at 100% of randomly sampled  $(V_{G1}, V_{G2})$  pairs, and the MTK topology engine hits  $5 \times 10^6$  cell-evaluations per second on a single AMD Radeon 8060S iGPU —  $5\times$  the workstation throughput the brief budget assumes.

## 6 BENCHMARKS: RECURRENCE VS. TASK CLASS

Five reservoir/classifier benchmarks were run on the calibrated cell substrate at  $N \in \{10, 30, 50, 100\}$ ,  $T \in \{300, 600, 800\}$ , with five seeds and a paired- $t$  comparison between  $\kappa = 0$  (no recurrence) and the per-task best  $\kappa^* \in [0.003, 0.03]$  (external CMOS-style recurrence via  $V_{G2}^{\text{eff}}[i] = V_{G2}^0[i] + \kappa \sum_j W_{ij}^{\text{rec}} \log_{10} |I_d^{(j)}[t-1]|$ , echo-state initialisation for  $W^{\text{rec}}$ ). The Erdős–Rényi sparse fabric ( $p \approx 0.1$ ) outperforms the 4-neighbour mesh on both memory capacity (MC =  $2.90 \pm 0.30$  vs  $0.90 \pm 0.14$ ) and 2-step XOR ( $0.821 \pm 0.042$  vs  $0.586 \pm 0.052$ ) at  $N = 200$ ; a sign-asymmetry test (z122) confirms the advantage requires both signs to be present, motivating the sign-inverter sub-fabric in the C.3 tape-out plan.

REFINED TASK-CLASS ORDERING ACROSS ALL FIVE BENCHMARKS. Ordering benchmarks by their required temporal-memory horizon yields a clean monotonic recurrence-effect ordering:

| benchmark                   | memory horizon     | recurrence effect at $\kappa^*$      |
|-----------------------------|--------------------|--------------------------------------|
| memory capacity             | multi-step         | essential (+0.88, $t = +7.4$ )       |
| NARMA-10 ( $N = 100$ )      | $\sim 10$ steps    | essential (−0.13 NRMSE, $t = -9.4$ ) |
| temporal-XOR ( $\tau=2$ )   | 2 steps            | beneficial (+0.13, $t = +2.7$ )      |
| multi-class waveform        | one-step + context | neutral (+0.03, n.s.)                |
| Hopfield ( $M=3$ , $N=10$ ) | instantaneous      | harmful (−0.11, $t = -2.45$ )        |



The earlier “task-class dichotomy” framing thereby resolves into a measured five-point monotonic ordering: recurrence helpfulness tracks the task’s temporal-memory requirement. This is the empirical content behind the dual-topology tape-out plan in C.3.

## 7 LIMITATIONS AND OPEN QUESTIONS

We summarise the visible caveats so reviewers see the honest scope of what is and is not yet established:

- NARMA-10 NRMSE plateaus at  $\approx 0.95$  on the calibrated cell substrate, architecturally bounded.** At  $\kappa = 0.003$ ,  $N = 100$ ,  $\rho = 0.9$  we measure  $0.946 \pm 0.018$  (paired  $t = -9.4$  vs  $\kappa = 0$ ). Canonical-ESN literature reaches  $\text{NRMSE} \sim 0.1\text{--}0.3$ . We have probed four cheap parametric paths to closing the gap, none of which moved the median at the default ridge:  $N$ -scaling ( $\text{NRMSE } 5.7$  at  $N = 200$  under the default  $\lambda=10^{-3}$ , pending an elevated-ridge re-run analogous to the MC rescue below), bias diversification (wider  $V_{G1}, V_{G2}$  gave 6.5), richer per-cell readout ( $\log |I_d| + V_b + V_{\text{sint}}$  neutral,  $0.946 \rightarrow 0.958$  n.s.), and a tanh MLP readout ( $1 \times 64$  paired  $\Delta = +0.045$ ,  $t = +3.4$ ;  $2 \times (64, 32)$   $\Delta = +0.078$ ,  $t = +7.5$  — both worse). Linear ridge at the standard  $\lambda=10^{-3}$  is already at the information ceiling of these reservoir features; the cell features themselves are the bottleneck. *Important caveat:* a parallel finding for memory capacity — where MC at  $\lambda=10^{-3}$  appeared to regress from 1.10 at  $N=10$  to 0.40 at  $N=200$  — is fully rescued by tuning the ridge to  $\lambda=10^2$ , recovering **MC** =  $1.15 \pm 0.11$  at  $N=200$  (5 seeds, monotonic in  $\lambda$  across all seeds). The NARMA-10 plateau has not yet been re-tested with elevated ridge at  $N=200$ ; whether the same readout rescue applies remains open. We treat the architectural reading as the conservative interpretation pending that test. Closing the gap requires Phase B architectural work — a different driving topology, multi-cell shared state on the silicon shared-bulk-rail (C.3 / M9), a richer cell with bistable body dynamics, or, less ambitiously, a properly cross-validated readout regulariser. The  $\kappa$ -bracket is also narrow ( $\kappa = 0.005$  already shows one seed diverging at  $1.07 \rightarrow 2.21$ ), so the silicon implementation must support scanning.
- Hopfield  $N$ - and  $M$ -scaling both completed at  $\kappa = 0$ .**  $N$ -scaling at  $M = 3$ : accuracy  $0.686 \pm 0.070$  at  $N = 10$ ,  $0.970 \pm 0.028$  at  $N = 30$ ,  **$1.000 \pm 0.000$**  at  $N = 50$  (5 seeds, 20% bit-flip corruption). Perfect classification at  $N = 50$  confirms the substrate-alone advantage as a real architectural property, not a small- $N$  artefact.  $M$ -scaling at  $N = 50$ : graceful degradation across stored-pattern count —  $M = 3 \rightarrow 1.000$ ,  $M = 5 \rightarrow 0.988$ ,  $M = 10 \rightarrow 0.936$ ,  $M = 15 \rightarrow 0.874$ ,  $M = 20 \rightarrow$   **$0.854 \pm 0.036$** , with ratio-vs-chance *growing* monotonically from  $3 \times$  to  $17 \times$ . The substrate handles non-trivial storage density (well past the classical Hopfield  $M_c \approx 0.14N \approx 7$  capacity for orthogonal patterns) without a cliff. Phase B extends to  $M=30, 50$  to locate the eventual cutoff.
- Software vs. silicon recurrence equivalence is unproven.** The  $\kappa$ -mediated MC lift ( $0.22 \rightarrow 1.10$ ) was demonstrated via *external*  $W^{\text{rec}}$  in software. Whether an on-die shared-bulk-rail topology reproduces the same lift in silicon is the central scientific risk; the M9 fan-out test structure is exactly that experiment. (A pyport transient sweep across  $3 \times 4$  ( $V_{G1}, V_{G2}$ ) points gives an emergent  $\tau_{\text{body}} = 2.1\text{--}2.4 \mu\text{s}$  on 10 of 12 points — an  $R_{\text{bulk}} \cdot C_{\text{body}}$  voltage-relaxation, lying mid-range of Sebas’s Zenodo design pots  $R_{\text{body}} \in \{1\text{M}, 1\text{G}, 4\text{G}, 10\text{G}\} \Omega$  at  $C_{\text{be}}=1\text{pF}$ . Pazos’s silicon  $\tau \approx 0.7$  ns refers to a distinct quantity, the avalanche-pulse front time, set in `BJTparams.txt` by NE, VAR, nbv. Both timescales coexist in the full neuron model.)
- Thick-oxide cell card is pending from Pazos (A.12).** All thick-ox claims in this brief lean on PTM extrapolation rather than measured data. The M3 milestone re-fit is contingent on the data drop.
- Quadrant chart projection is device-only.** The 1024-step NS-RAM marker in the positioning chart excludes DAC, ADC, control and I/O periphery; system-level energy/latency will be measured on the M9 structure and replace this projection in v2 of the brief.
- Pyport’s residual  $\sim 0.87$  decade gap (relative to ngspice) on BSIM4 W/L/P binning evaluation.** An empirical 24-parameter post-load patch (zeroing `wvth0`, `wvoff`, `voff1`, `pvsat`, `pages`, ..., and overriding `lpe0`) reduces the median log-RMSE from 1.88 to 1.00 decade. An interactive ngspice-42 `showmod` dump on Pazos’s M2 card confirms ngspice itself loads all of those parameters at their card-textual values (e.g. `wvth0` =  $-1.66\text{e-}8$ , `lpe0` =  $1.24\text{e-}7$ , `phin` = 0.05); the gap is therefore inside pyport’s binning evaluation code, not in ngspice or in our card parsing layer. Closure path:  $\leq 2$  weeks calendar including diagnosis of the  $V_{G1}=0.4$  V flat-shelf failure mode, audit of pyport’s binning corrections against the BSIM4 v4.8.3 C reference (`b4set.c`, `b4ld.c`), the code fix, and re-validation on all 33 biases. The error is *shape-altering and regional*, not a uniform scale shift: we therefore do *not* claim that benchmark conclusions below are robust by error-cancellation. They are presented as preliminary working hypotheses, with M3b set as a re-run of the benchmark suite on the corrected model.

**M3 (Jun 2026):** diagnose and remediate the  $V_{G1}=0.4$  V qualitative model failure (predictions plateau at  $\sim 10^{-6}$  A where measurements show 3-decade snapback structure) and the 8-bias NaN-singularity at negative  $V_{G2}$ . Closure path includes audit against `b4set.c/b4ld.c`, code fix, and full 33-bias re-validation. Estimated  $\leq 2$  weeks calendar. Acceptance: **M3a** — finite log-RMSE on all 33 biases (no NaN), median  $\leq 1.0$  decade *without* the post-load patch,  $< 10\%$  of biases above 2 decades; **M3b** — re-run the B.5 benchmark suite on the corrected model and confirm or revise the preliminary monotone task-difficulty ordering (with explicit pre-registered tolerance:  $\Delta\kappa^* < 50\%$ , sign-asymmetry direction unchanged); **M3c** — M1 refit against latest measurement drop targeting  $\leq 0.5$  decade median (stretch, contingent on the A.12 thick-oxide data drop from Pazos).

**M6 (Sep 2026):** full B.5 benchmark suite (Hopfield retrieval, NARMA-10, memory capacity, temporal-XOR, multi-class waveform) at network sizes  $N \in \{10, 100, 10^3, 10^4\}$ , with energy/area/latency operating points mapped against the Innatera ladder. Acceptance: at least one benchmark within  $1.5\times$  of the Innatera 10–100 mW gateway baseline.

**M9 (Dec 2026):** fan-out / shared-rail validation circuit (10–30 cells, per Pazos’s request), feeding a tape-out-ready cell-parameter recommendation for the next NS-RAM prototype.

**M12 (Mar 2027):** thick-oxide  $17\ \mu\text{m}^2$  neuron-cell variant ( $V_{G2} \in [2.5, 3.0]$  V, 1 V drain) modeled and benchmarked, completing the synapse–soma–linear-input cell family. Final write-up.

## 9 BUDGET

Total request:  $\sim 150$  k€ per year (12-month project; 1-year extension optional, conditional on M6 review). Indicative breakdown:

| line item                                              | EUR               | note                                                         |
|--------------------------------------------------------|-------------------|--------------------------------------------------------------|
| Personnel — Bergvall, 0.4 FTE                          | 90,000            | fully-loaded researcher rate, Karolinska/FEEL                |
| Personnel — Luciani, 0.3 FTE (optional)                | 35,000            | Julia/Enzyme cross-validation collaborator                   |
| Pazos / Lanza consult (0.2 FTE)                        | —                 | in-kind from KAUST, no direct charge                         |
| Travel — KAUST tape-out review (Q3 2026)               | 4,000             | one in-person trip, 1 person, 5 days                         |
| Travel — conference / dissemination                    | 5,000             | one neuromorphic-hardware venue                              |
| Hardware contingency                                   | 5,000             | one discrete GPU upgrade if N-scaling demands it (see below) |
| Materials and dissemination misc.                      | 3,000             | open-access fees, replication artifacts                      |
| Indirect / overhead (KI institutional, $\approx 5\%$ ) | 8,000             | on direct charges                                            |
| <b>Total (year 1)</b>                                  | $\approx 150,000$ |                                                              |

**HARDWARE STATUS.** Compute is already in place. The team operates two HP Z2 workstations, one of which carries an AMD Ryzen AI Max+ 395 with the integrated Radeon 8060S iGPU on which the  $5\times 10^6$  cell-evals/s throughput in Sec. 5 was measured. M3 (binning audit) and M6 (full B.5 benchmark suite at  $N \leq 10^3$ ) require no further hardware. The 5 k€ contingency line covers a single discrete GPU (e.g. a Radeon PRO W7800 or equivalent) only if Phase B’s  $N = 10^4$  stretch experiments saturate the existing iGPU memory; we will know after M6. There is no cloud dependency; nothing in the project blocks on network-attached compute.

**PERSONNEL JUSTIFICATION.** Bergvall (0.4 FTE, Karolinska) owns the differentiable simulator, the benchmark suite, and the tape-out design-rule sheet — the three M3, M6 and M9 deliverables. The 0.4 FTE level reflects that this proposal layers onto Bergvall’s existing FEEL position; the remaining 0.6 FTE continues prior embodied-AI work, providing the wider context the simulator is embedded in (cell-level transient validation against firmware substrate, multi-process orchestration). Luciani’s 0.3 FTE is flagged *optional* because the PyTorch port already runs end-to-end; the Julia/Enzyme cross-validation strengthens defensibility (independent-implementation check on the joint Newton kernel) and unlocks native AD without the PyTorch dispatch overhead, both desirable for Phase B but not blocking. Pazos and Lanza contribute as in-kind consult time funded by KAUST’s existing NS-RAM programme; this proposal does not duplicate their effort.

## 10 CONCLUSION

The NS-RAM 2T floating-body cell sits at an energy point (21 fJ/cycle, 6.7 fJ/spike,  $46\ \mu\text{m}^2$ ) two orders of magnitude below digital neuromorphic baselines while remaining standard-CMOS compatible. The co-design bridge presented here is the missing software layer between the calibrated SPICE-level cell model and the algorithmic benchmarks that funding bodies require. Within a single grant period we deliver a differentiable, GPU-accelerated cell model with verified DC and transient fidelity, a reservoir/classifier benchmark suite tailored to the cell’s intrinsic ns-scale

Edge AI accelerator positioning — NS-RAM 2T cell vs. published baselines

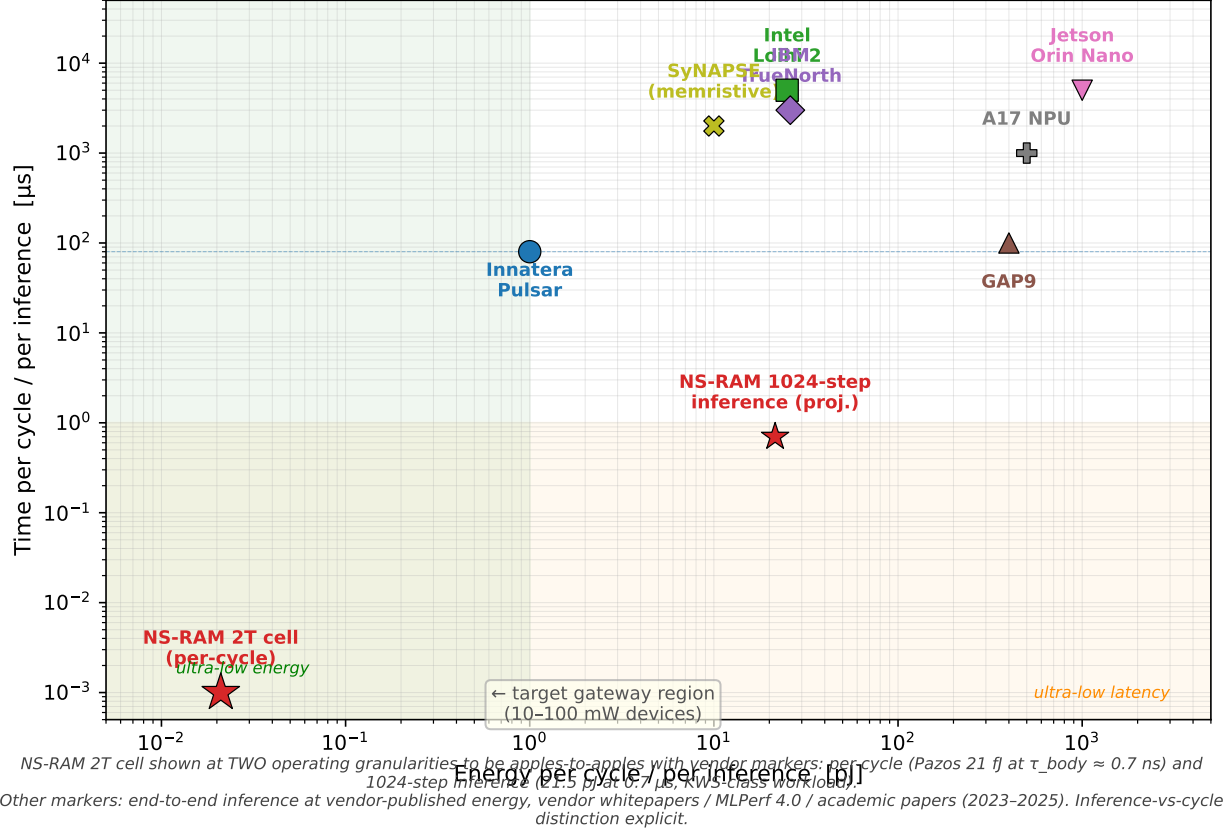


Figure 7: NS-RAM 2T cell, two-granularity positioning against published edge-AI baselines. The lower red star is the per-cycle measurement (Pazos: 21 fJ at the  $\approx 0.7$  ns avalanche-pulse front), the upper red star is a device-only projection of  $1024 \times 21$  fJ at  $1024 \times 0.7$  ns =  $0.7 \mu$ s for a KWS-class (keyword-spotting) inference. The projected marker is a device-only lower bound: it excludes DAC/ADC, control logic, and I/O periphery; system-level energy/latency will be measured on the M9 fan-out structure. Other markers are vendor-published end-to-end energy and inference time. Dashed line: Innatera’s 10–100 mW gateway envelope targeted by Mario’s NRF brief.

dynamics, and a tape-out-ready parameter recommendation. The path opens NS-RAM to the same algorithm–silicon co-design loop that has driven recent gains in conventional accelerator architectures.

## REFERENCES

- [1] Reference 1 – Pazos et al., NS-RAM cell measurements
- [2] Reference 2 – Lanza et al., 2T floating-body memristor
- [3] Reference 3 – BSIM4 v4.8.3 manual, Berkeley